Bashir Agro Admin Panel
Phase 3: Farm Location Management Module

Scope

Implement only Farm Location Management after Phase 2.
Do not implement Flutter app, REST API, live GPS tracking, mobile geofence validation, Task Management, or future API integration.

Objective

Convert the existing Farm Location section into a fully dynamic Laravel + PostgreSQL module while preserving the existing Bengali green agricultural UI, tables, forms, cards, modals and responsive layout.

Database Table: farm_locations

Columns:

id
location_code (unique, generated server-side)
name
type
address
area
area_unit
latitude
longitude
geofence_radius_meters
status
created_by
created_at
updated_at
deleted_at

Use soft deletes.

Database Table: farmer_location

Pivot table for farmer and location relationship.

id
farmer_id
farm_location_id
created_at
updated_at

Relationship:
One farmer can have many locations.
One location can have many farmers.

Models

Create:

app/Models/FarmLocation.php

Relationships:

FarmLocation belongsTo creator(User)
FarmLocation belongsToMany Farmer

Update Farmer model:

Farmer belongsToMany FarmLocation.

Location Code Generation

Generate automatically from backend.

Example:

LOC-000001
LOC-000002
LOC-000003

Admin must not manually enter location codes.

Controller

Create:

app/Http/Controllers/Admin/FarmLocationController.php

Methods:

index()
create()
store()
show()
edit()
update()
destroy()

Keep controllers clean and use Eloquent.

Form Requests

Create:

StoreLocationRequest.php
UpdateLocationRequest.php

Validate:

name
type
area
area_unit
latitude
longitude
geofence_radius_meters
status

Create Location Workflow

Admin submits location form.

Flow:

Blade Form
↓
Laravel Validation
↓
Generate Location Code
↓
Save PostgreSQL Record
↓
Create Activity Log
↓
Redirect

Use DB::transaction().

Location List

Database-driven table should show:

Location Code
Name
Type
Area
Assigned Farmers
Radius
Status
Actions

Remove all hard-coded location data from JavaScript.

Search and Filters

Search:

location name
location code
type
address

Filters:

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■■■■■■■
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Support combined query:

/admin/locations?search=value&status;=active

Pagination

Use Laravel pagination.

Recommended:

20 locations per page.

Maintain filters using withQueryString().

Farmer Assignment

Admin can attach farmers to locations.

Example:

North Field:
- Rahim
- Karim

Store relationship inside:

farmer_location

Removing assignment only removes pivot record,
not the farmer.

Dashboard Integration

Replace hard-coded active location numbers.

Example:

FarmLocation::where('status','active')->count()

Dashboard values must come from PostgreSQL.

Activity Logs

Create logs:

location.created
location.updated
location.deleted
location.assigned
location.unassigned

Store:

user_id
action
subject_type
subject_id
old_values
new_values
ip_address

Security

Use:

Authentication middleware
CSRF protection
Form Requests

Authorization
Mass assignment protection
Soft deletes

Testing

Create:

tests/Feature/FarmLocationManagementTest.php

Test:

- Admin access
- Create location
- Location code generation
- Validation failure
- Update location
- Soft delete
- Search
- Status filters
- Pagination
- Farmer attach
- Farmer detach

Antigravity Prompt

```
Proceed to PHASE 3 ONLY: Farm Location Management Module.

Use the existing Bashir Agro Laravel project after Phase 2.

Do not redesign the UI.

Do not implement:
- Flutter API
- REST API
- GPS tracking
- Mobile geofence validation
- Task Management

Implement:

- farm_locations table
- farmer_location pivot table
- FarmLocation model
- Farmer relationship update
- FarmLocationController CRUD
- StoreLocationRequest
- UpdateLocationRequest
- Location code generation
- Search
- Filters
- Pagination
- Farmer assignment
- Activity logging
- Feature tests

Preserve Bengali UI, green theme, cards, forms and tables.
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Remove hard-coded location data from JavaScript.

Database must be the source of truth.

Run migrations.
Run tests.
Report files created and modified.

STOP after Phase 3.
Do not start Phase 4 automatically.
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